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Analysis of short-term treatment with the phosphodiesterase type 5 inhibitor tadalafil on long bone development in young rats

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²Department of Orthopaedics, Shandong University Qilu Hospital, Shandong University, Jinan, China; ³Department of Orthopaedics, The First Affiliated Hospital of the Medical College, Shihezi University, Shihezi, China; ⁴Department of Pediatrics, Perelman School of Medicine, University of Pennsylvania, Philadelphia, Pennsylvania; and ⁵Division of Endocrinology and Diabetes and the Center for Bone Health, The Children's Hospital of Philadelphia, Philadelphia, Pennsylvania

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Wang L, Jia H, Tower RJ, Levine MA, Qin L. Analysis of short-term treatment with the phosphodiesterase type 5 inhibitor tadalafil on long bone development in young rats. *Am J Physiol Endocrinol Metab* 315: E446–E453, 2018. First published June 19, 2018; doi:10.1152/ajpendo.00130.2018.—Cyclic GMP (cGMP) is an important intracellular regulator of endochondral bone growth and skeletal remodeling. Tadalafil, an inhibitor of the phosphodiesterase (PDE) type 5 (PDE5) that specifically hydrolyzes cGMP, is increasingly used to treat children with pulmonary arterial hypertension (PAH), but the effect of tadalafil on bone growth and strength has not been previously investigated. In this study, we first analyzed the expression of transcripts encoding PDEs in primary cultures of chondrocytes from newborn rat epiphyses. We detected robust expression of PDE5 as the major phosphodiesterase hydrolyzing cGMP. Time-course experiments showed that C-type natriuretic peptide increased intracellular levels of cGMP in primary chondrocytes with a peak at 2 min, and in the presence of tadalafil the peak level of intracellular cGMP was 37% greater ($P < 0.01$) and the decline was significantly attenuated. Next, we treated 1-mo-old Sprague Dawley rats with vehicle or tadalafil for 3 wk. Although $10 \text{ mg} \cdot \text{kg}^{-1} \cdot \text{day}^{-1}$ tadalafil led to a significant 52% ($P < 0.01$) increase in tissue levels of cGMP and a 9% reduction ($P < 0.01$) in bodyweight gain, it did not alter long bone length, cortical or trabecular bone properties, and histological features. In conclusion, our results indicate that PDE5 is highly expressed in growth plate chondrocytes, and short-term tadalafil treatment of growing rats at doses comparable to those used in children with PAH has neither obvious beneficial effect on long bone growth nor any observable adverse effect on growth plate structure and trabecular and cortical bone structure.

bone; endochondral ossification; growth plate; PDE5 inhibitor; young rat

INTRODUCTION

In mammals, development and growth of skeletal long bones occur through a process of endochondral ossification in which the skeletal cartilage anlagen are replaced by mineralized bone tissue. Postnatally, endochondral ossification at the growth

plate is essential for longitudinal bone growth (36). During this process, chondrocytes proliferate and undergo hypertrophy and apoptosis. A cartilage template is synthesized by the dividing chondrocytes, which is later invaded by blood vessels that bring osteoclasts that resorb the matrix. In turn, osteoblasts, the bone-forming cells, are recruited and deposit bone matrix on remnants of the cartilage model. With maturation of the skeleton and closure of epiphyseal growth plates, longitudinal growth of bone ceases. Nevertheless, skeletal remodeling persists throughout life as part of a continuous process of bone replacement and repair that involves a coordinated interplay between osteoclastic bone resorption and osteoblastic bone formation.

Multiple intracellular signals are necessary to regulate cartilage development and bone remodeling, and recently much interest has been focused on the role of intracellular cyclic GMP (cGMP), generated by both soluble and particulate forms of guanylate cyclase that are stimulated by nitric oxide and C-type natriuretic peptide (CNP), respectively (43, 48). Like cyclic AMP (cAMP), cGMP is an intracellular second messenger that mediates the action of various hormones and chemicals on a large variety of cells. In bone, proliferation of osteoblasts can be increased by mechanical stimulation via a mechanism that is cGMP-dependent and associated with nitric oxide-induced stimulation of soluble guanylate cyclase (sGC; 40). Interestingly, at least part of the anabolic effect of estrogen on osteoblasts appears to be mediated via this same pathway (24, 25, 29). For example, cinaciguat, a small molecule agonist of sGC, can reverse the effects of estrogen deficiency on bone, producing improved trabecular bone microarchitecture in ovariectomized animals with effect sizes similar to those obtained with estrogen (25). Similarly, osteoblasts and chondrocytes express the transmembrane receptor protein NPR2, a particulate guanylate cyclase that is activated upon binding of its primary ligand CNP (34, 39, 44). In chondrocytes, CNP-induced cGMP antagonizes FGFR3 signaling and thereby regulates the rate of chondrocyte hypertrophy (37). Genetic mutations that increase NPR2-dependent cGMP elevations in chondrocytes, either through increased serum CNP levels (8) or gain-of-function missense mutations of NPR2 (17, 30, 31), cause skeletal overgrowth in humans and mice (23). In addition, in mice the loss of *Npr3*, a decoy receptor for CNP

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and therefore an inhibitor for CNP action or overexpression of brain natriuretic peptide, another ligand for Npr2 in mice, also result in skeletal overgrowth (23, 45). By contrast, in humans inactivating mutations in one or both *NPR2* alleles lead to proportionate decreases in synthesis of cGMP in patients with short stature or acromesomelic dysplasia (type Maroteaux dwarfism), respectively (4, 6, 13, 35, 49, 51, 52). Moreover, deletion of either the *Cnp* or *Npr2* gene in mice results in severe dwarfism caused by reduced postnatal endochondral ossification (10, 49).

These examples provide compelling evidence for a concentration-dependent effect of cGMP on long bone growth and bone remodeling and suggest that pharmacological manipulation of chondrocyte cGMP levels might have important effects on skeletal development and statural growth. In children, agents that inhibit phosphodiesterase (PDE) type 5 (PDE5), which selectively hydrolyzes cGMP, are commonly used to treat pulmonary arterial hypertension (PAH), a disease associated with impaired growth, particularly in younger children (38). Approximately 6% to 18% of extremely low-birth weight infants (birth weight less than 1,000 g) may develop signs of PAH (3), with a median age of diagnosis of 112 days of life (7). Incidence data from the Netherlands have revealed an annual incidence and point prevalence of 0.7 and 4.4 for idiopathic PAH and 2.2 and 15.6 for PAH related to congenital heart disease cases per million children (50). Initial oral therapy usually consists of pulmonary vasodilators, including endothelin receptor antagonists or more commonly a PDE5 inhibitor (32) such as tadalafil (47) or sildenafil (5), which can increase levels of cGMP in vascular cells. There is greater enthusiasm for oral use of tadalafil ($0.5\text{--}1\text{ mg}\cdot\text{kg}^{-1}\cdot\text{day}^{-1}$) in children based on studies showing increased clinical efficacy and safety and improved compliance in children with PAH compared with sildenafil (42, 47). However, because of the importance of cGMP in skeletal development, a possible effect of tadalafil on bone growth in these young patients remains a theoretical concern. In this study, we first analyzed *Pde5* expression profile in epiphyseal chondrocytes from newborn rats and then performed a comprehensive analysis of skeletal development in young rats receiving tadalafil to identify potential effect(s) of this PDE5 inhibitor on bone growth.

METHODS

Animals. Male Sprague Dawley rats, purchased from Charles River, were grouped randomly and treated with the following reagents via oral gavage daily starting from 1 mo of age: vehicle (0.5% methyl cellulose; cat. no. M-638, Sigma-Aldrich, St. Louis, MO, $2.5\text{ mg}\cdot\text{kg}^{-1}\cdot\text{day}^{-1}$ tadalafil, and $10\text{ mg}\cdot\text{kg}^{-1}\cdot\text{day}^{-1}$ tadalafil (Lilly USA LLC, Indianapolis, IN) ($n = 10$ rats/group). Animals were weighed each week and euthanized 3 wk later for harvest of the long bone and brain. Additional 1-mo-old rats were treated with 10 mg/kg tadalafil and euthanized right before the treatment or at 3, 6, and 24 h after treatment ($n = 3$ /time point) for harvest of the brain. In accordance with the standards for animal housing, rats were group-housed at $23^{\circ}\text{C}\text{--}25^{\circ}\text{C}$ with a 12-h light/dark cycle and allowed free access to water and standard laboratory pellets. All animal work performed in this report was approved by the Institutional Animal Care and Use Committee at the University of Pennsylvania. The study design and organization respected the Animal Research: Reporting of In Vivo Experiments guidelines.

Primary chondrocyte culture and real-time RT-PCR. Primary rat chondrocytes from newborn pups were harvested from the knee

epiphyseal region and cultured as described previously (55). Cells were plated at a density of $4 \times 10^4/\text{cm}^2$ in 6-well plates in growth medium (DMEM containing 15% FBS, $100\text{ }\mu\text{g/ml}$ streptomycin, and 100 U/ml penicillin). RNA was harvested when cells reached 90% confluence for qRT-PCR of PDEs using primers listed in Table 1.

Measurement of cGMP. The rat cerebrum was dissected out and homogenized. Primary chondrocytes were cultured to near confluence and then treated with 100 nM recombinant CNP (cat. no. N-8768, Sigma-Aldrich) in the absence or presence of $50\text{ }\mu\text{M}$ tadalafil (cat. no. SML-1877, Sigma-Aldrich) and $250\text{ }\mu\text{M}$ IBMX (cat. no. I-5879, Sigma-Aldrich). The cGMP concentrations in brain or cell culture were quantified by an ELISA immunoassay kit (cat. no. ADI-900-164, Enzo Life Sciences, Farmingdale, NY) following the manufacturer's instructions.

Microcomputed tomography analysis. Rats were anesthetized by isoflurane on day 0, day 7, day 14, and day 21 for measuring the long bone growth. To do so, their right tibiae were secured by a customized holder to ensure minimal motion effect before they were scanned by an in vivo microcomputed tomography (microCT) system (VivaCT 40, Scanco Medical AG). A scout view of the entire tibia was performed to measure the length of the tibia from the upper extremity to the lower extremity.

After euthanasia, the right tibiae were harvested for trabecular bone analysis using VivaCT 40. The proximal end of the tibia corresponding to a 0- to 4.4-mm region below the lowest point of growth plate was scanned at $10.5\text{-}\mu\text{m}$ isotropic voxel size. All images were smoothed by a Gaussian filter ($\sigma = 1.2$, support = 2.0) and then thresholded corresponding to 579 mg HA/cm^3 . The images of the secondary spongiosa regions located at 1.5–3.0 mm below the lowest point of the growth plate were contoured for trabecular bone analysis. Trabecular volumetric bone mineral density (BMD), bone volume fraction, trabecular thickness, trabecular separation, trabecular number, and structure model index were calculated by three-dimensional standard microstructural analysis (9).

For cortical bone analysis, the right femurs were harvested and scanned via scout view to determine the midline (the central region with the same distance to both ends of femurs). The midshaft region (from 0.5 mm above to 0.5 mm below the midline) was scanned at $10.5\text{-}\mu\text{m}$ isotropic voxel size. All images were smoothed by a Gaussian filter ($\sigma = 1.2$, support = 2.0) and then thresholded

Table 1. Sequences of rat primers used for RT-PCR

Gene	5' Primer	3' Primer
<i>β-actin</i>	CGCGAGTACAACCTTGC	CGCAGCGATATCGTCATCCA
<i>Pde1a</i>	CCACTTTGTGATCGGAAGTC	TTCTGCTGAATGATGTCCACC
<i>Pde1b</i>	CAGGGTGACAAGGAGGCAGAG	GACATCTGGTTGGTGATGCC
<i>Pde1c</i>	TCTCAAAGGATGACTGGAGG	GCTTCTCTGCACCGTGTC
<i>Pde2a</i>	CCTCTGTGACCTCTCTGACC	TGAACCTGTGGGACACCTTGG
<i>Pde3a</i>	TCACAGGGCCTTAACCTACAC	GGAGCAAGAATTGGTTTGTC
<i>Pde3b</i>	CCTCAGGCAGTTTATACAAATG	TGCTTCTCATCTCCCTGCCTC
<i>Pde4a</i>	GTGGAGAAGTCTCAGGTGGG	TGGAACCTGTCAGGCAAGG
<i>Pde4b</i>	TAGAAGATAACAGGAAGTGG	GCAATGTCTATGTGAGTCTC
<i>Pde4c</i>	ACGTGGCGTACCACAACAGC	TACCGCGAGGTGATGGTTCTC
<i>Pde4d</i>	GGATAATGGAGGAGTTCTTCC	CGATTTGCTCCAAAGTGTC
<i>Pde5a</i>	CCCTGGCCTATTCAACAACGG	GTGGGTCAAGGCCCTCATACAG
<i>Pde6a</i>	CCGAGATTGTCTTCCCTTTG	GCTATGCCCTGTGTTGTTG
<i>Pde6b</i>	TTCTCTGGTCAGCCAATAAG	GAGTACCGGTACAGTTTAG
<i>Pde6c</i>	CCAGTTTCTTGGATGGTCTC	GGCTTTGGTGAGGTTTCATA
<i>Pde7a</i>	TGGACAAGCCAAGTGTATGCTG	TTTAAGTAACAGTGCATGGCC
<i>Pde7b</i>	AAAGCCCAAGTGGGAAGAGC	GCAAGGGAGCTGGTAAATG
<i>Pde8a</i>	CAACAAGCCTCTGAAAGC	TGGTCTGGGAGAAATAC
<i>Pde8b</i>	ACCACAACCTCCACCATG	AGAGGCTTGTGTGATGCTG
<i>Pde9a</i>	TGGGTGGACTGTTTACTGG	CGGTCTTCATTGCTTTTCG
<i>Pde10a</i>	TCGCTGGTGGCGTTGTGATG	TTCTCTGATGCTGGGATGTAC
<i>Pde11a</i>	TTTAGCGGTGATTGTGGG	TCTCGAAGTACAGCGTGAGG

corresponding to 643.1 mg HA/cm³. Cortical thickness, cortical tissue mineral density, polar moment of inertia, cortical area, periosteal perimeter, and endosteal perimeter were measured using the microCT scanner software (Scanco Medical).

Histomorphometry. The right tibiae were fixed in 4% PFA overnight, washed under tap water, and then decalcified in 0.5 M EDTA (pH 7.4) for 3 wk before embedding in paraffin. Six-micrometer-thick sections were cut for H&E staining. The left tibiae were processed for embedding in methylmethacrylate. Six-micrometer-thick longitudinal sections were cut with a Polycut-S motorized microtome (Reichert) and stained with Goldner's trichrome solutions. Images corresponding to the secondary spongiosa region were acquired via a Nikon Eclipse 90i and quantified using Bioquant Osteo Software (Bioquant Image Analysis). Osteoblast number per bone surface and osteoclast number per bone surface were calculated as described earlier (11). The thicknesses of the proliferative zone and hypertrophic zone of the growth plate were expressed as the average of 12 measurements that were taken at sites that were evenly distributed across the entire growth plate in the paraffin sections as described previously (55). The proliferative zone is defined as the area that contains flattened cells in a longitudinal and columnar arrangement. The hypertrophic zone is defined as the area at which the cell size starts to increase to the area where cartilage becomes bone.

Statistical analysis. Data are expressed as means $\pm$ standard error (SE) and analyzed by unpaired Student's *t*-test or with a one-way or two-way ANOVA with Bonferroni's post hoc test for multiple comparisons using Prism 5 software (GraphPad Software). Values of *P* < 0.05 were considered significant.

RESULTS

Tadalafil enhances the cGMP response to CNP in primary chondrocytes. Previous studies have demonstrated the expression of PDE5 in human articular cartilage, a chondrogenic cell line (12), and an osteoblastic cell line (2). However, the expression profile of all PDE isoforms and their activities in epiphyseal chondrocytes, particularly PDE5, were still largely unknown. To determine which PDEs play a primary role in targeting cGMP signaling pathway in chondrocytes, we first measured mRNA levels of all PDEs, including those that specifically hydrolyze cAMP (*Pde4*, *Pde7*, and *Pde8*), those that specifically hydrolyze cGMP (*Pde5*, *Pde6*, and *Pde9*), and those that hydrolyze both (*Pde1*, *Pde2*, *Pde3*, *Pde10*, and *Pde11*), in primary chondrocytes derived from the epiphyseal

region of rat long bones. Among those PDEs, we detected the expression of *Pde1a*, *1c*, *2a*, *3a*, *4a*, *4b*, *4d*, *5a*, *7a*, *7b*, *8a*, and *10a* (Fig. 1A). Therefore, *Pde5* is the only cGMP-specific PDE expressed in primary chondrocytes. Although chondrocytes also express PDEs that hydrolyze both cGMP and cAMP, such as *1a*, *1c*, *2a*, *3a*, and *10a*, their expression levels were 100 times lower compared with that of *Pde5*.

Next, we investigated whether PDE5 regulates cGMP levels in chondrocytes after CNP stimulation. Time course experiments revealed that CNP rapidly and transiently increases the intracellular cGMP level in primary chondrocytes with a peak at 2 min and a rapid fall in cGMP thereafter (Fig. 1B). Combined treatment of cells with CNP plus either tadalafil, which specifically inhibits PDE5, or IBMX, which inhibits all PDEs, not only significantly elevated the peak level of intracellular cGMP but also attenuated the subsequent decline of cGMP compared with CNP alone. Overall, IBMX showed a stronger effect of elevating cGMP level at all time points than tadalafil. These results suggest that PDE5-dependent hydrolysis of cGMP plays an important role in cGMP homeostasis in chondrocytes but other PDEs may also be involved in cGMP hydrolysis.

In vivo treatment of tadalafil increases tissue cGMP level in rats. To study whether tadalafil affects skeletal development in vivo, we treated 1-mo-old male rats with vehicle, 2.5 mg/kg or 10 mg/kg tadalafil daily by oral gavage for 3 wk. Those dosages attempt to mimic clinical treatment of children with PAH at 0.5–1 mg·kg⁻¹·day⁻¹. Both doses significantly decreased the gain of body weight during this period (Fig. 2A). At the end of treatment, we validated the efficacy of tadalafil by measuring the tissue level of cGMP. ELISA analysis revealed that although 2.5 mg·kg⁻¹·day⁻¹ of tadalafil treatment does not alter cGMP level, 10 mg·kg⁻¹·day⁻¹ of tadalafil treatment causes a significant 52% increase in brain cGMP compared with vehicle only (Fig. 2B). Furthermore, rats given a single injection of 10 mg tadalafil showed that tissue cGMP levels were 2.0-fold greater at 3 h than at baseline, with a peak increase of 2.9-fold around 6 h. These data are consistent with the role of tadalafil as a potent inhibitor of PDE5 activity. Taken together, our results demonstrate that our tadalafil treat-

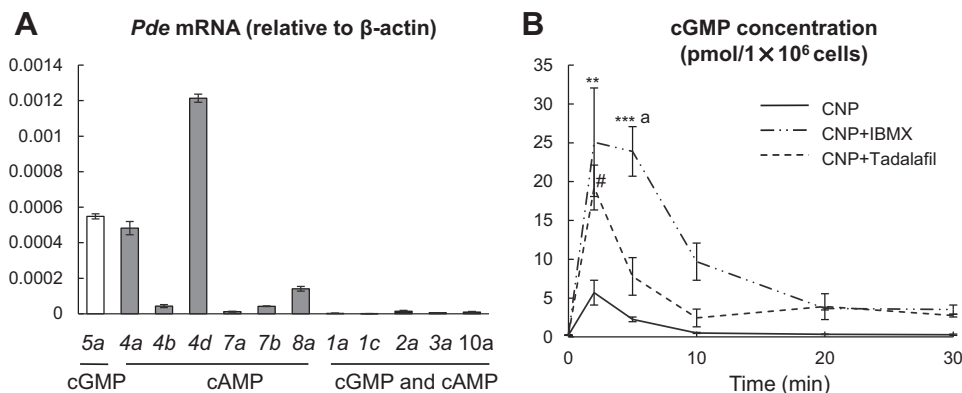


Fig. 1. Tadalafil enhances the cGMP response to CNP in primary chondrocytes. A: qRT-PCR analysis of *Pde* isoforms in primary chondrocytes. *n* = 3 rats. B: time-courses of CNP-stimulated intracellular cGMP levels in primary chondrocytes with or without pretreatment of IBMX (250 μ M) or tadalafil (50 μ M). Cells were collected from indicated time points after CNP (100 nM) addition and analyzed by ELISA to measure intracellular cGMP amounts (*n* = 3). #*P* < 0.05, CNP + tadalafil vs. CNP; ***P* < 0.01, CNP + IBMX vs. CNP; ****P* < 0.001, CNP + IBMX vs. CNP; ^a*P* < 0.05, CNP + IBMX vs. CNP + tadalafil by two-way ANOVA with Bonferroni's post hoc test for multiple comparisons. cGMP, cyclic GMP; CNP, C-type natriuretic peptide; qRT-PCR, quantitative RT-PCR.

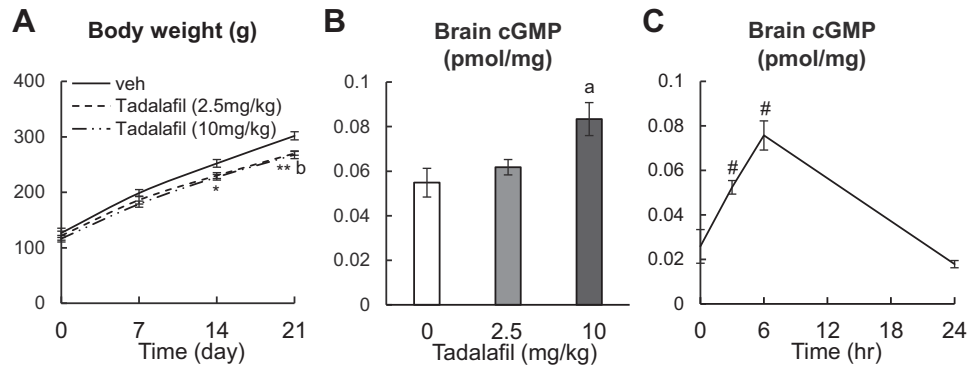


Fig. 2. Tadalafil treatment increases the tissue cGMP level in young rats. *A*: rats were weighed weekly during the 3-wk tadalafil treatment period. $n = 10$ rats/group. *B*: ELISA analysis of cGMP amount in brain tissue of rats treated with vehicle, 2.5 mg/kg or 10 mg/kg tadalafil daily for 3 wk. $n = 5$ rats/group. *C*: ELISA analysis of cGMP amount in brain tissue of rats at 0, 3, 6, and 24 h after receiving 1 dose of 10 mg/kg tadalafil. $n = 3$ rats/time point. $*P < 0.05$, 2.5 mg/kg tadalafil vs. vehicle; $**P < 0.01$, 2.5 mg/kg tadalafil vs. vehicle; $^aP < 0.05$, 10 mg/kg tadalafil vs. vehicle; $^bP < 0.01$, 10 mg/kg tadalafil vs. vehicle; $^{\#}P < 0.05$ vs. 0 h by two-way (*A*) or one-way ANOVA (*B* and *C*) with Bonferroni's post hoc test for multiple comparisons. cGMP, cyclic GMP; veh, vehicle.

ment regimens are effective in increasing tissue levels of cGMP level in young rats and they produce a known pharmacological effect on body weight.

Tadalafil treatment does not affect longitudinal bone growth. To examine the possible effect of tadalafil on the endochondral bone formation, we first measured tibial growth in rats during the treatment period by weekly in vivo microCT scans. As shown in Fig. 3, *A* and *B*, the longitudinal elongation of tibiae was not affected by tadalafil when given

at either 2.5 or 10 mg/kg per day. Three weeks later, rats were euthanized, and their tibiae were harvested for histology assessment. Surprisingly, the structure of the growth plate was very similar in vehicle and tadalafil groups (Fig. 3*C*). Quantification did not detect any change in the thicknesses of the proliferative zone and hypertrophic zone of the growth plate (Fig. 3*D*). The vascular invasion at the chondral-osseous junction remained the same after 3 wk of tadalafil treatment (Fig. 3*E*). Moreover, we did not observe

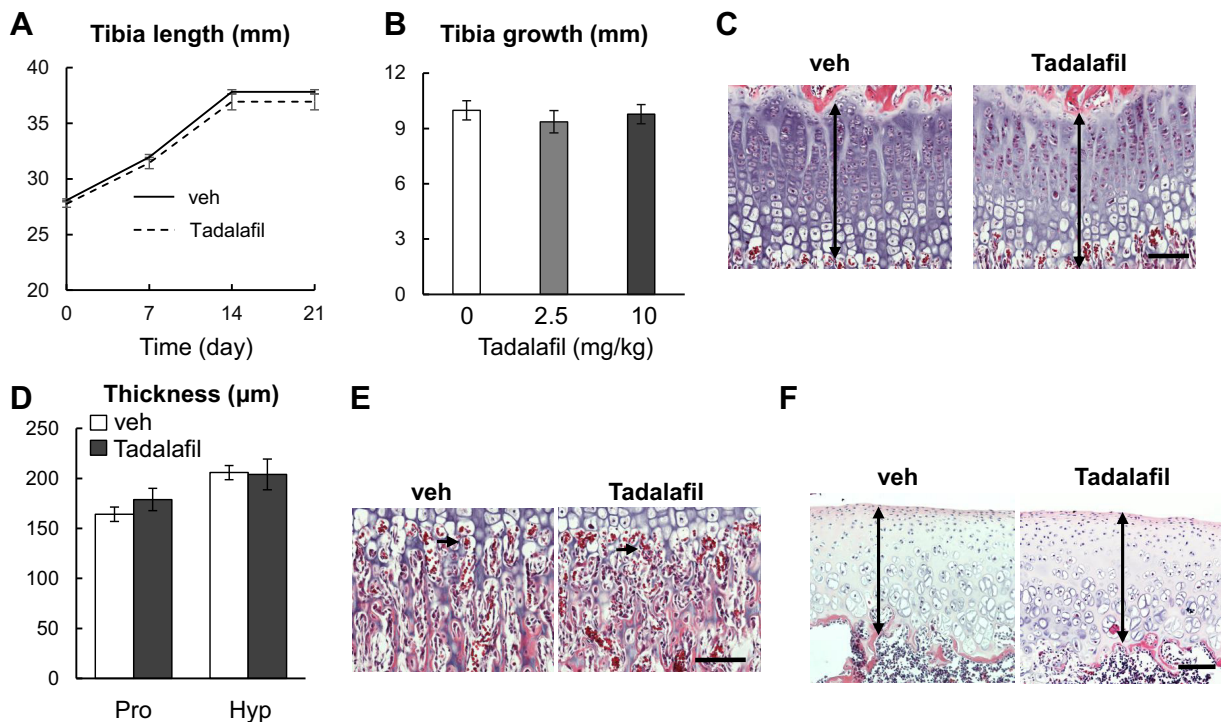


Fig. 3. Three weeks of tadalafil treatment does not affect longitudinal skeletal growth. *A*: tibia length of vehicle-treated and 10 mg·kg⁻¹·day⁻¹ tadalafil-treated rats was measured weekly by in vivo microcomputed tomography. $n = 10$ rats/group. *B*: length of tibia in rats after 3 wk of vehicle, 2 mg·kg⁻¹·day⁻¹, and 10 mg·kg⁻¹·day⁻¹ tadalafil treatments. $n = 10$ rats/group. *C*: representative H&E staining of growth plate from vehicle group and 10 mg·kg⁻¹·day⁻¹ tadalafil group. Double-headed lines point to the growth plate. *D*: thickness of the proliferative zone (Pro) and hypertrophic zone (Hyp) of the growth plate was quantified. $n = 5$ rats/group. *E*: H&E staining of chondral-ossification junction of vehicle-treated and 10 mg·kg⁻¹·day⁻¹ tadalafil-treated rats. Arrows point to erythrocytes within blood vessels that invade growth plate cartilage. *F*: H&E staining of tibial articular cartilage of vehicle-treated and 10 mg·kg⁻¹·day⁻¹ tadalafil-treated rats. Double-headed lines point to the articular cartilage. Scale bar: 100 μm. Results were analyzed by two-way (*A*, *D*) or one-way (*B*) ANOVA with Bonferroni's post hoc test for multiple comparisons. veh, vehicle.

any abnormality in the knee articular cartilage (Fig. 3F). Therefore, our data suggested that tadalafil treatment *in vivo* does not influence chondrocyte proliferation, hypertrophy, and cartilage remodeling into bone during endochondral ossification.

Tadalafil treatment does not alter trabecular bone structure or bone cell metabolism. Growth plate cartilage is eventually replaced by bone to form trabecular bone in the primary and secondary spongiosa. MicroCT analysis of the proximal tibial metaphysis indicated that all trabecular bone structural parameters (BMD, bone volume fraction, trabecular number, trabecular thickness, trabecular separation, and structure model index) are the same among vehicle, 2.5 mg·kg⁻¹·day⁻¹, and 10 mg·kg⁻¹·day⁻¹ tadalafil treatment groups (Fig. 4, A and B). Furthermore, bone histomorphometric analyses indicated that there are no significant changes in osteoblast and osteoclast numbers in the trabecular bone (Fig. 4C), suggesting that tadalafil treatment does not alter either bone formation or bone resorption *in vivo*.

Tadalafil treatment did not alter cortical bone structure. Cortical bone is derived from perichondrium surrounding the epiphyseal cartilage during endochondral ossification. To characterize the effects of tadalafil treatment on cortical bone, we scanned femoral midshaft region at the end of treatment. Similar to trabecular bone, we did not observe any significant changes in cortical bone structural parameters, such as endosteal perimeter, periosteal perimeter, cortical BMD, cortical thickness, cortical area, and polar moment of inertia after 3 wk of tadalafil treatment (Fig. 5, A and B).

DISCUSSION

Recent guidelines from the American Heart Association (AHA) and American Thoracic Society (ATS) recommend the use of PDE5 inhibitors, including tadalafil, for the treatment of pediatric PAH (1). Although no adverse (or beneficial) effects of tadalafil on patient skeletons have been reported to date, the low prevalence of pediatric PAH, and the untoward consequences of PAH itself on normal growth, make it difficult to assess a potential clinical effect of these agents on skeletal development and statural growth. This uncertainty prompted the current study in which we used adolescent rats to investigate whether tadalafil treatment might be associated with changes in growth plate development and long bone structure. To our knowledge, our cell culture studies are the first comprehensive analysis of PDE isoforms expressed in chondrocytes and our rat experiments are the first animal study to validate the clinical observation of tadalafil effects on the skeleton.

Although tadalafil enhanced CNP-induced accumulation of cGMP in chondrocytes *in vitro*, which might predict altered endochondral bone growth (17, 30, 40, 41) or remodeling (21) *in vivo*, we found no evidence for a skeletal effect of tadalafil in young rats using doses (2.5 and 10 mg·kg⁻¹·day⁻¹) that are comparable to or slightly in excess of the AHA/ATS-suggested dose of 0.5–1 mg·kg⁻¹·day⁻¹ for pediatric patients (33). By contrast, our tadalafil treatments were effective in blunting weight gain in growing rats, an effect that has been ascribed to the ability of increased cGMP to enhance energy expenditure by inducing differentiation of brown adipocytes and “brown-

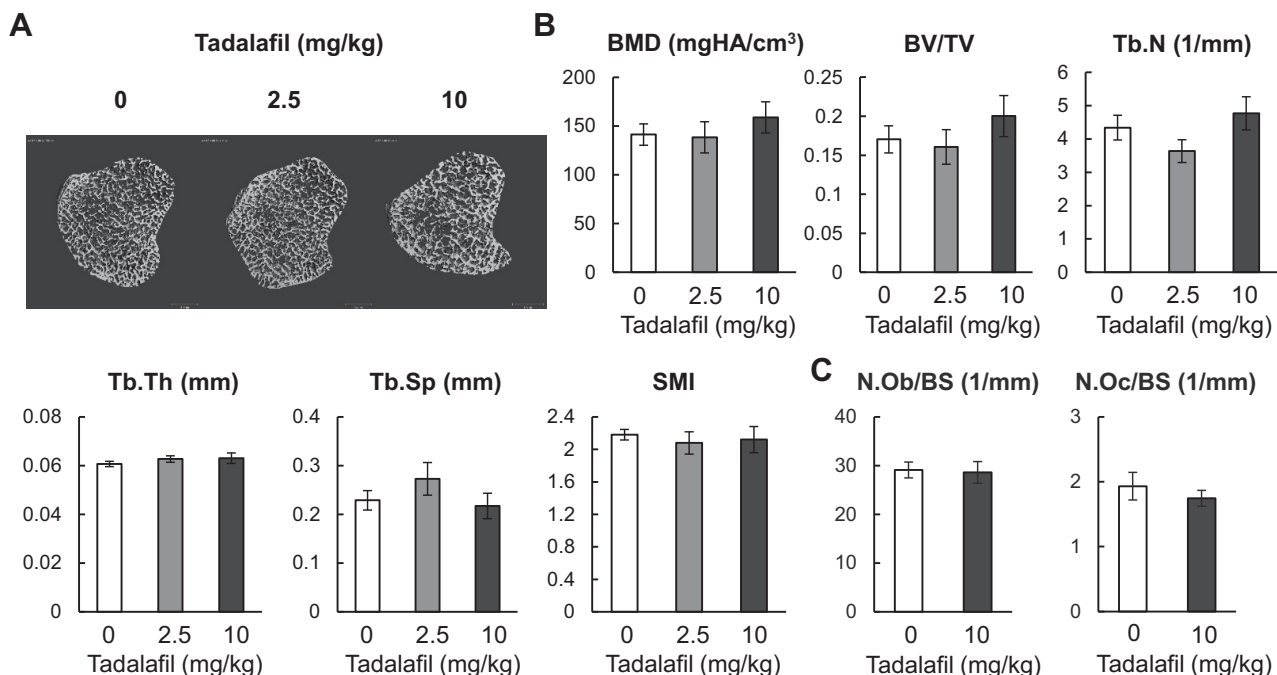


Fig. 4. Three weeks of tadalafil treatment does not change bone metabolism in the trabecular bone. A: representative 3-D reconstructed microcomputed tomography (microCT) images of tibial trabecular bone in rats treated with vehicle, 2.5 mg·kg⁻¹·day⁻¹ or 10 mg·kg⁻¹·day⁻¹ tadalafil. B: microCT measurement of trabecular bone structural parameters. *n* = 10 rats/group. C: osteoblast numbers (N.Ob) and osteoclast numbers (N.Oc) per bone surface (BS) were quantified by histomorphometric analysis. *n* = 5 rats/group. Results were analyzed by one-way ANOVA with Bonferroni's post hoc test for multiple comparisons (B) or by unpaired *t*-test (C). 3-D, three-dimensional; BMD, bone mineral density; BV/TV, bone volume fraction; SMI, structure model index; Tb.N, trabecular number; Tb.Sp, trabecular separation; Tb.Th, trabecular thickness.

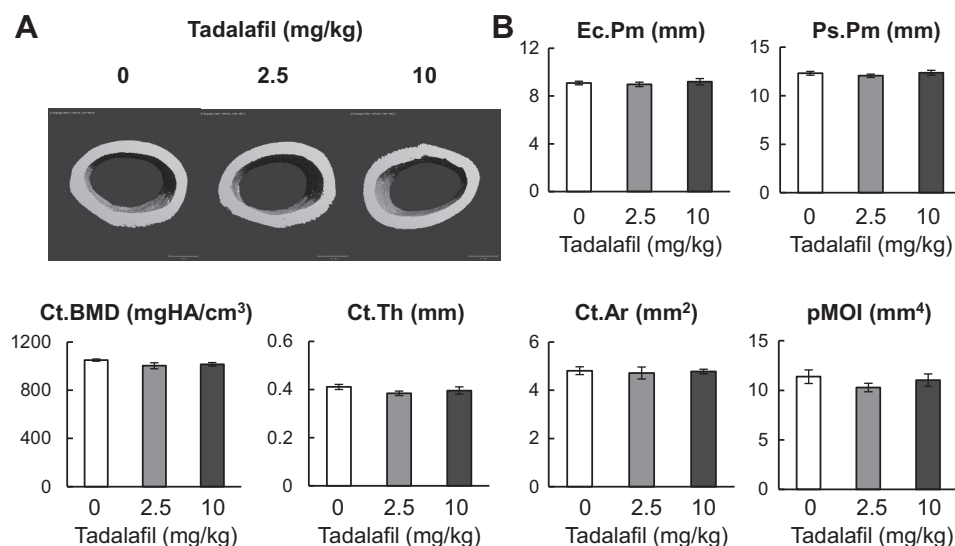


Fig. 5. Cortical bone structure is not altered by 3 wk of tadalafil treatment. *A*: representative 3-D reconstructed microcomputed tomography (microCT) images of cortical bone in femoral midshaft region of rats treated with vehicle, 2.5 mg·kg⁻¹·day⁻¹ or 10 mg·kg⁻¹·day⁻¹ tadalafil. *B*: microCT measurement of cortical bone structural parameters. *n* = 10 rats/group. Results were analyzed by one-way ANOVA with Bonferroni's post hoc test for multiple comparisons. Ct.Ar, cortical area; Ct.BMD, cortical tissue bone mineral density; Ct.Th, cortical thickness; Ec.Pm, endosteal perimeter; pMOI, polar moment of inertia; Ps.Pm, periosteal perimeter.

ing" of primary white adipocytes (14, 18, 19, 53), thus providing evidence that our dose regimen was capable of producing a pharmacological effect. We envision several possible explanations to account for the inability of therapeutic doses of tadalafil to affect skeletal development. First, the dose of tadalafil that we administered may have been too low to achieve effective concentrations of the drug in growth plate chondrocytes because of the avascular nature of cartilage. In our experiments, we observed a dosage-dependent effect in terms of upregulation of concentration of cGMP in brain tissue, and as described above, achieved a recognized physiological end point of weight reduction. Although it is possible that higher doses of tadalafil might affect the skeleton, these doses would be in excess of currently prescribed amounts of tadalafil and therefore, of no clinical consequence. Second, it is possible that other PDEs, such as PDE1, 2, 3, and 10, which are expressed at much lower levels than PDE5 in primary chondrocytes, might exert significant effects *in vivo* and compensate for the tadalafil-induced loss of PDE5 activity in chondrocytes, thereby preventing an increase in intracellular cGMP. A previous study has showed that in ATDC5 chondrogenic cells (Mouse 129 teratocarcinoma AT805 derived), PDE1, a calcium-calmodulin stimulated PDE, plays an important role in cGMP hydrolysis (12). Consistent with this notion, we found that CNP-induced accumulation and maintenance of cGMP in primary cultures of chondrocytes is greater in the presence of IBMX, a generalized inhibitor of PDEs, than in the presence of tadalafil, which is a PDE5-specific inhibitor. Third, tadalafil may have increased cGMP levels in chondrocytes but not to an extent that would permit an alteration in cellular behavior or growth. Gain-of-function mutations that activate NPR2 and lead to skeletal overgrowth are associated with intracellular levels of cGMP that are 20-fold greater than normal (41), which are far greater than 4- to 5-fold increases we observed with either IBMX or tadalafil treatment in primary chondrocytes. Indeed, a separate study using riociguat, an allosteric stimulator of the sGC, to treat newborn rat pups generated a similar conclusion that drugs effective in treating PAH in newborn rat pups by elevating the cGMP level do not affect skeletal development *in vivo* (11a).

The CNP/Npr/cGMP signaling pathway regulates not only cartilage homeostasis but also bone metabolism. A recent study showed that overexpressing CNP in mice results in bone loss in intact femurs and accelerated fracture healing because of increased bone turnover (27). A series of studies at earlier time points demonstrated that in culture, CNP stimulates osteoblast proliferation, differentiation, and mineralization via elevating the cGMP level (16, 22, 28, 46, 54). However, another *in vivo* study found that tadalafil treatment in 2-mo-old mice for 2 mo reduces trabecular bone mass by suppressing Wnt-induced osteoblastogenesis (15). This study used extremely high doses of tadalafil (45 and 75 mg·kg⁻¹·day⁻¹), which greatly exceed the equivalent human doses, and which may have caused toxic effects. Nevertheless, no obvious growth plate phenotype was reported in this study. The role of CNP/Npr/cGMP signaling pathway in osteoclastogenesis is also ambiguous. Although an early report suggested that CNP increases osteoclast bone resorption in a mixture of bone marrow cells (20), another study identified an opposite and inhibitory action of CNP on osteoclast formation from bone marrow macrophages (26). Based on these prior studies, it is likely that our inability to demonstrate any changes in our microCT analyses of tibial trabecular bone and femoral cortical bone and static histomorphometry of tibiae in young rats treated with tadalafil for 3 wk represents a failure of tadalafil to induce a robust increase in intracellular cGMP in bone cells as high as CNP does to cause any alteration in either bone formation or resorption activity.

Our studies have several limitations. First, the rat model used in this study is at the adolescent stage. Pediatric PAH affects children as young as the new born. The skeletal responses to tadalafil might be age-dependent. Second, we only studied male rats, but pediatric PAH affects both women and men. Last, we exposed rats to tadalafil for only a short period of time. However, as the goal of our study was to determine the potential effects of tadalafil on children, we limited exposure of rats to tadalafil to only 3 wk, which corresponds to roughly 2 yr in humans. Although it might require a longer treatment period to identify tadalafil-related skeletal changes, we believe that such effects if detected should be minor and more related to adults instead of children.

In conclusion, our results provide a comprehensive gene expression profile of PDEs in rat cartilage chondrocytes. We also demonstrated that treatment with tadalafil for 3 wk at doses of up to 10 mg·kg⁻¹·day⁻¹, which are much higher than doses used in children with PAH, has no observable effect on long bone growth, including growth plate structure and trabecular and cortical bone structure, in growing rats. Therefore, our studies indicate that despite the important roles of cGMP in skeletal regulation at commonly used dosages, tadalafil is unlikely to affect the skeleton of growing children.

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DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the authors.

AUTHOR CONTRIBUTIONS

L.W., H.J., M.A.L., and L.Q. conceived and designed research; L.W., and H.J. performed experiments; L.W., H.J., R.J.T., and L.Q. analyzed data; L.W., H.J., M.A.L., and L.Q. interpreted results of experiments; L.W., and L.Q. prepared figures; L.Q. drafted manuscript; M.A.L. edited and revised manuscript; M.A.L. and L.Q. approved final version of manuscript.

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